PIPELINE AUDIT AND HEALTH REPORT

Logistics: e-commerce fulfilment

Orders are picked by three pickers (express orders jump the queue), then packed at two stations; 3% fail packing checks and go back to picking over a bypass lane. A sorter routes express orders to a courier dock and standard ones into 15-order truck loads that depart after 15 minutes at the latest. Packing is the constraint.

Generated 2026-10-13 19:48 | simulated 8.0 h (warm-up 30 min) | seed 44 | 10 replications

HEALTH SCORE

89 /100

Grade B

Good: sound design with minor weaknesses worth addressing.

Critical 0 | High 1 | Medium 0 | Low 1 | Info 2

The score starts at 100 and loses points for each finding. The deductions are listed in Appendix B.

THROUGHPUT	COMPLETED	ARRIVALS LOST
80.7 units/h	629 units	0.0%

AVERAGE WIP	CYCLE TIME	BOTTLENECK
14.9 units	680 s	pack_1

Executive summary

The line keeps up with demand (0.00% lost, sources never throttled): throughput equals the arrival rate, so nothing is binding right now. The busiest station is 'pack_1' at 96% busy; that is where capacity runs out first as demand grows.

What to fix first

- High (pack_1): Add parallel capacity, speed the station up, or reduce downtime.
- Low (failure model): Add realistic MTBF and MTTR to the stations that break down in practice.

Best measured improvement

No single local change moves throughput by more than 1%: the line is demand-limited or constrained elsewhere. Try raising the arrival rate, or look at the system-wide structure.

Findings

Each finding names where the problem is, why it matters, and what to do. Severity reflects the impact on throughput, correctness or confidence in the results.

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
High	Capacity	**pack_1**	The bottleneck is 96% utilised (busy or down). There is no slack: any disturbance directly cuts output. Action: Add parallel capacity, speed the station up, or reduce downtime.	-8
Low	Reliability	**failure model**	No machine has a failure model (MTBF/MTTR), so the results are optimistic. Action: Add realistic MTBF and MTTR to the stations that break down in practice.	-3

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
Info	Capacity	**pack_1**	Highest utilisation in the line (96%); the line currently keeps up with demand. Action: This is where capacity runs out first as demand grows.	
Info	Reliability	**dependencies**	2 stations lie on every path (serial structure), but none has a failure model. Action: Fine for an ideal analysis; add failure data to find out how fragile this is.	

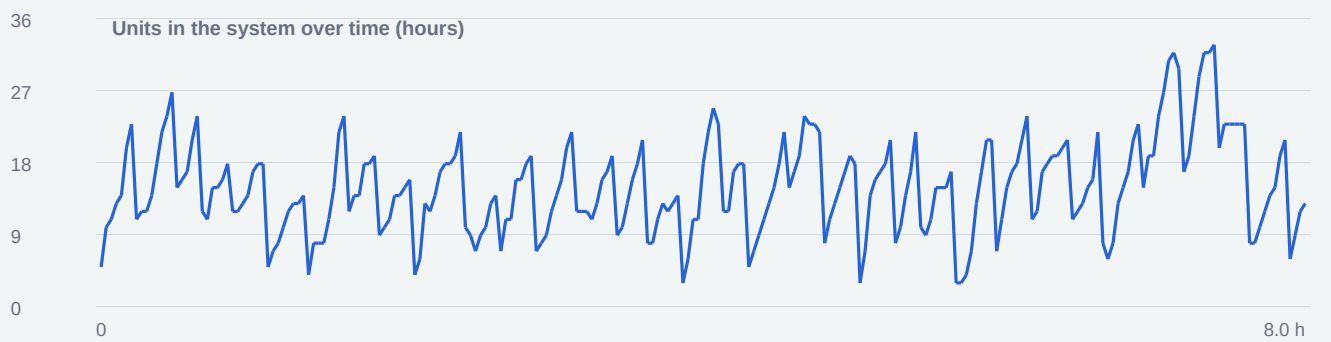
Performance by node

Share of the measured window each node spent busy, blocked (finished part cannot leave), down (failed) and starved (waiting for work).

Legend: green = busy, amber = blocked, red = down, grey = starved or idle.

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
orders	source	-	-	-	-	-
picker_1	machine		79.2	3.7	0.0	1.3 / 5
picker_2	machine		42.5	3.7	0.0	0.7 / 4
picker_3	machine		14.1	3.4	0.0	0.3 / 3
pack_1 bottleneck	machine		96.2	0.0	0.0	2.5 / 5
pack_2	machine		76.1	0.0	0.0	1.9 / 5
labeler	machine		18.2	0.0	0.0	0.2 / 2
sorter	machine		6.7	0.0	0.0	0.1 / 1
express_dock	machine		16.8	0.0	0.0	0.2 / 2
standard_batch	batcher		6.9	0.0	0.0	7.9 / 17
shipped	sink	-	-	-	-	-

Work in process over time



Reliability and robustness

The pipeline was simulated 10 times with different random seeds. A wide interval means the result depends strongly on luck.

METRIC	MEAN	95% CONFIDENCE	STD DEV	RELATIVE
Throughput	80.03 units/h	+- 2.35	3.29	2.9%
Average WIP	13.91 units	+- 0.67	0.93	4.8%
Cycle time	630.95 s	+- 18.40	25.72	2.9%
Arrivals lost	0.00 %	+- 0.00	0.00	0.0%

No replication deadlocked.

Failure model

No machine has a failure model.

Single points of failure

All units must pass through: labeler, sorter. If one of these stops, output stops.

Model validity checks

CHECK	RESULT	DETAIL
Unit conservation	PASS	created 645 = completed 629 + scrapped 0 + lost 0 + in system 16
Little's Law	PASS	L = 14.94, lambda x W = 15.24 (gap 2.0%)
Run length	PASS	window = 40 x cycle time (30x or more recommended)
Warm-up	PASS	first 30 min excluded from statistics
Deadlock-free	PASS	reference run completed

What-if analysis

Each lever was applied to one of the most constrained stations and the whole pipeline re-simulated over at least 8 hours with 3 identical random seeds per scenario. Baseline throughput: 80.6 units/h.

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
pack_1	+2 buffer slots	80.8	**+0.2**	+0.2%
pack_2	+2 buffer slots	80.8	**+0.2**	+0.2%
picker_1	10% faster processing	80.8	**+0.2**	+0.2%
picker_1	+2 buffer slots	80.6	**+0.0**	+0.0%
pack_2	10% faster processing	80.5	**-0.1**	-0.2%
pack_1	10% faster processing	80.4	**-0.3**	-0.3%

No single local change moves throughput by more than 1%: the line is demand-limited or constrained elsewhere. Try raising the arrival rate, or look at the system-wide structure.

Appendix A: node inventory

Effective parameters after node-type templates are applied. Parameters not listed use the node type's defaults.

NODE	TYPE	EFFECTIVE PARAMETERS
orders	source	interarrival=exponential(45), mix={"express":0.25,"standard":0.75}, priority={"express":2}
picker_1	machine	capacity=8, discipline=priority, process=normal(60,15)
picker_2	machine	capacity=8, discipline=priority, process=normal(60,15)
picker_3	machine	capacity=8, discipline=priority, process=normal(60,15)
pack_1	machine	capacity=4, discipline=priority, process=normal(75,15)
pack_2	machine	capacity=4, discipline=priority, process=normal(75,15)
labeler	machine	capacity=5, process=normal(8,2)
sorter	machine	capacity=5, process=3
express_dock	machine	capacity=4, process=30
standard_batch	batcher	batch_size=15, capacity=30, process=60, timeout=900
shipped	sink	(defaults)

Appendix B: how the score is computed

Health score = 100 minus the sum of the penalties below, never less than 0. Grades: A 90 and above, B 80 to 89, C 70 to 79, D 60 to 69, F below 60. Penalties are capped per category so that one repeated issue cannot dominate the score. Informational findings carry no penalty.

SEVERITY	WHERE	FINDING	POINTS
High	**pack_1**	The bottleneck is 96% utilised (busy or down). There is no slack: any disturbance directly cuts output.	-8
Low	**failure model**	No machine has a failure model (MTBF/MTTR), so the results are optimistic.	-3
	Total deductions		**-11**
	Health score		**89**

Method: the pipeline is simulated as a discrete-event system; the bottleneck index is utilisation (busy plus down time) with small weights for starved downstream and blocked upstream neighbours. What-if results are measured, not estimated. Confidence intervals use Student's t over independent replications.